

1. Administrative & Study Identification

Full Study Title: A Phase 1, First-in-human, Multicentre, Open-label, Dose Escalation Study of [225Ac]-FPI-2068 in Adult Patients With Advanced Solid Tumours **Short Title/Acronym:** A Phase 1, Dose-escalation Study of [225Ac]-FPI-2068 in Adult Patients With Advanced Solid Tumours **Protocol Number:** FPI-2068-101 (AZD2068) **NCT Number:** NCT06147037 **Sponsor Name:** AstraZeneca **Investigational Product:** [225Ac]-FPI-2068 (Actinium-225 conjugated to FPI-2053), [111In]-FPI-2107, and FPI-2053 **Phase of Development:** Phase I **Medical Monitor:** AstraZeneca Clinical Operations Lead / Lead Physician: Aditya Juloori

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of [225Ac]-FPI-2068, determine the optimal pre-dose administration of FPI-2053, and establish the Recommended Phase 2 Dose (RP2D). **Secondary Objectives:** To evaluate the preliminary anti-tumor activity (efficacy) and investigate the dosimetry, biodistribution, and pharmacokinetics (PK) of [225Ac]-FPI-2068, [111In]-FPI-2107, and FPI-2053.

2.2 Background & Rationale To address the critical unmet medical need in patients with advanced solid tumors (such as Head and Neck Squamous Cell Carcinoma [HNSCC], Non-Small Cell Lung Cancer [NSCLC], metastatic Colorectal Cancer [mCRC], Pancreatic Ductal Adenocarcinoma [PDAC], Gastric Cancer [GC], and Renal Cell Carcinoma [RCC]) whose disease has progressed despite prior treatment. [225Ac]-FPI-2068 is an EGFR-cMET targeted radiopharmaceutical therapy delivering an alpha emitter (actinium-225) directly to tumor cells.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Dose-escalation) **Blinding:** Open-label **Randomization:** N/A (Non-randomized)

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 70 to 110 adult participants **Number of Sites:** Multicenter (Global sites including US centers such as University of Chicago, MD Anderson, Fred Hutch, Stanford) **Estimated Study Start:** 31-Jul-2024 **Estimated Primary Completion:** 12-May-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years. 2. Histologically and/or cytologically confirmed solid tumor that is metastatic, locally advanced, recurrent or inoperable. Disease must have progressed despite prior treatment, with no standard therapy available, tolerated, or accepted by the participant. 3. Measurable disease as defined by RECIST Version 1.1. 4. ECOG Performance Status of 0 or 1.

4.2 Exclusion Criteria 1. Previous treatment with any systemic radiopharmaceutical. 2. Prior anti-cancer therapy without adequate washout and recovery from toxicities. 3. Radiation therapy (RT) within 28 days prior to the first dose of [111In]-FPI-2107. 4. Uncontrolled pleural effusion, pericardial effusion, or ascites requiring recurrent drainage procedures ($\geq$ once per month). 5. Patients with known Central Nervous System (CNS) metastatic disease unless treated and stable.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: - **Part A:** Optimization of FPI-2053 dose administered pre-dose with a fixed dose of [225Ac]-FPI-2068. - **Part B:** Dose escalation of [225Ac]-FPI-2068 using the optimal dose of FPI-2053 established in Part A. Participants receive dosing every 56 days. The study also utilizes a fixed dose of the imaging agent [111In]-FPI-2107 prior to treatment. **Route of Administration:** Intravenous (IV) **Duration of Treatment:** Up to 3 cycles of the Treatment Period (approx. 24 weeks).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for managing underlying disease symptoms. **Prohibited:** Concurrent anti-cancer therapies, systemic radiopharmaceuticals, or radiation therapy lacking the protocol-defined 28-day washout.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	-28 to 0	Informed Consent, Eligibility Assessment, Baseline RECIST Imaging
2	Day 1	Administration of [111In]-FPI-2107 imaging agent, dosimetry scans, baseline PK
3	Every 8 Weeks	Intravenous dosing of [225Ac]-FPI-2068 (every 56 days), RECIST v1.1 Tumor Assessments ($\approx$ 1 week)
4	Up to 5 Years	Long-term follow-up every 3 months ($\approx$ 2 weeks) for 2 years, then every 6 months for 3 years, or until disease progression

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Safety and tolerability assessments to determine Dose-Limiting Toxicities (DLTs), the Maximum Tolerated Dose (MTD), and Recommended Phase 2 Dose (RP2D). **Measurement Method:** Incidence and severity of Adverse Events (AEs) and Serious Adverse Events (SAEs) graded per CTCAE.

7.2 Secondary & Exploratory Endpoints - Anti-tumor activity (Objective Response Rate, Progression-Free Survival) evaluated via RECIST v1.1. - Pharmacokinetics (PK): Measurement of clearance, AUC, Cmax, and half-life of [225Ac]-FPI-2068, [111In]-FPI-2107, and FPI-2053. - Radiation dosimetry estimates to the whole body, organs, and regions of interest.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring from the first dose until 56 days after the last dose of investigational product. **Serious Adverse Events (SAEs):** Routine 24-hour reporting protocols for severe toxicities related to alpha-emitter radiation. **Data Monitoring Committee (DMC):** An internal Safety Review Committee will review safety and dosimetry data from each cohort to make dose-escalation decisions for Part B.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients receiving ≥ 1 dose of the investigational product); Efficacy/Response-Evaluable Set. **Sample Size Justification:** $N = 70$ to 110 is adequate for a Phase 1 open-label dose-escalation study to determine MTD and define safety/PK profiles across multiple solid tumor cohorts. **Handling of Missing Data:** Standard Phase 1 protocols apply; dose-limiting toxicity (DLT) evaluations require complete observation windows unless missingness is due to drug-related toxicity.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Rigorous onsite and remote monitoring to ensure protocol adherence, focusing on safe handling, administration, and disposition of radiopharmaceuticals. **Audit Trail:** Standard electronic data capture (eCRF) with detailed logs of dosimetry calculations and central imaging reads.

11. Study Identifiers & Governance

Primary ID: {{FPI-2068-101}} **Registry Number:** {{NCT06147037}}
Sponsor/Lead Organization: {{AstraZeneca}} **Study Title:** {{A Phase 1, Dose-escalation Study of [225Ac]-FPI-2068 in Adult Patients With Advanced Solid Tumours}} **Official Regulatory Title:** {{A Phase 1, First-in-human, Multicentre, Open-label, Dose Escalation Study of [225Ac]-FPI-2068 in Adult Patients With Advanced Solid Tumours}}

12. Clinical Framework (Radiopharmaceutical Specific)

Primary Condition: {{Advanced Solid Tumors}} **Diagnosis Threshold:** {{Refractory/metastatic progression despite standard therapy}}
Medical Methodology: {{Targeted Alpha-Emitter Radiopharmaceutical Therapy}} **Optimization Target:** {{EGFR-cMET targeted cell destruction using Actinium-225}}

13. Study Procedures & Methodology

Management Pathway: {{Dose escalation mapping optimal FPI-2053 pretreatment with fixed/escalating [225Ac]-FPI-2068}} **Education Protocol (HCP):** {{Radiation safety, safe handling of Ac-225 and In-111, administration protocols}} **Education Protocol (Patient):** {{Radiation exposure limits, post-dose hygiene precautions, and follow-up adherence}} **Quality Audit Mechanism:** {{Central dosimetry review and RECIST v1.1 central radiological confirmation}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to 5 years post-treatment}} **Onsite Visit Cadence:** {{Every 8 weeks during the 3-cycle treatment phase}}
Remote Monitoring Frequency: {{Every 3 months for 2 years, then every 6 months for 3 years}} **Checklist Review Interval:** {{Prior to each 56-day cycle administration}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				